Chapter test

27 Molecular genetics

Name: _____ ( ) Class: _____ Date: _____

Total marks: 40 marks
Time allowed: 40 minutes
Answer ALL questions.

Score: / 40

A. Concept checking (10 marks, 1 mark each)

1. The codons CGU and AGA both code for the amino acid arginine. What property of the genetic code does this illustrate?

   A. One gene codes for one protein.
   B. It is universal.
   C. It is degenerate.
   D. It is non-overlapping. ☐

2. Where do transcription and translation occur in eukaryotic cells?

	Transcription	***Translation***
A.	nucleus	nucleus
B.	nucleus	cytoplasm
C.	cytoplasm	nucleus
D.	cytoplasm	cytoplasm ☐

3. Which of the following biomolecules are involved in transcription?

   (1) DNA
   (2) RNA
   (3) amino acid

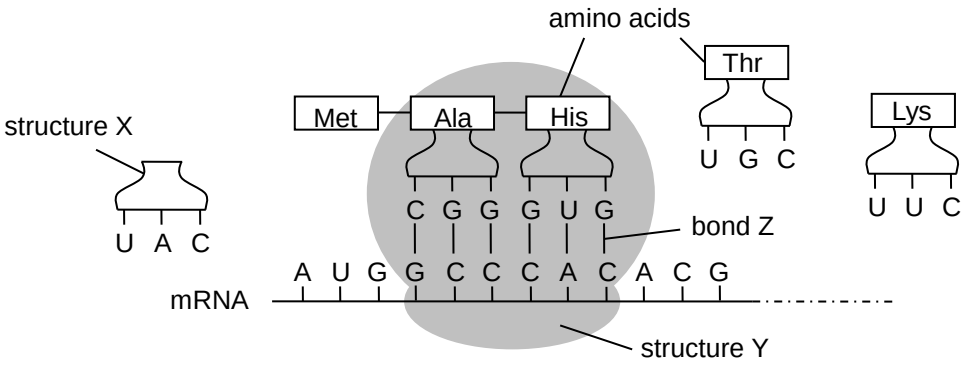
   A. (1) and (2) only
   B. (1) and (3) only
   C. (2) and (3) only
   D. (1), (2) and (3) ☐

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4. Which of the following descriptions about the process of translation is correct?

A. A new DNA strand is synthesized using an existing strand as a template.
B. Free RNA nucleotides bind to complementary bases on DNA.
C. Anticodons on mRNA bind to complementary codons on DNA.
D. Anticodons on tRNA bind to complementary codons on mRNA. ☐

Directions: Questions 5 and 6 refer to the diagram below which shows the process of protein synthesis.



5. Which of the following correctly identify X, Y and Z?

	X	***Y***	***Z***
A.	ribosome	enzyme	hydrogen bond
B.	tRNA	ribosome	hydrogen bond
C.	tRNA	enzyme	peptide bone
D.	tRNA	ribosome	peptide bone ☐

6. What is the mRNA codon for the amino acid Lys?

A. UUC
B. TTC
C. AAG
D. ACG ☐

★ 7. The tRNA anticodon for the sequence ATG on the coding strand of DNA is

A. UAC.
B. AUG.
C. TAC.
D. ATG. ☐

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8. Which of the follow descriptions about a mutation is/are correct?

(1) It is a product of natural selection.
(2) It is a change in an allele due to a change in DNA.
(3) It is caused by the random fusion of gametes.

A. (1) only
B. (2) only
C. (2) and (3) only
D. (1), (2) and (3) ☐

9. A mutation results in a change in the sequence of bases in a DNA molecule, as illustrated below.

Normal DNA: ATAAGCATGACATTA

Mutated DNA: ATAGAGCATGACATTA

What is this type of mutation?

A. insertion
B. deletion
C. substitution
D. inversion ☐

10. What is the cause of Down syndrome?

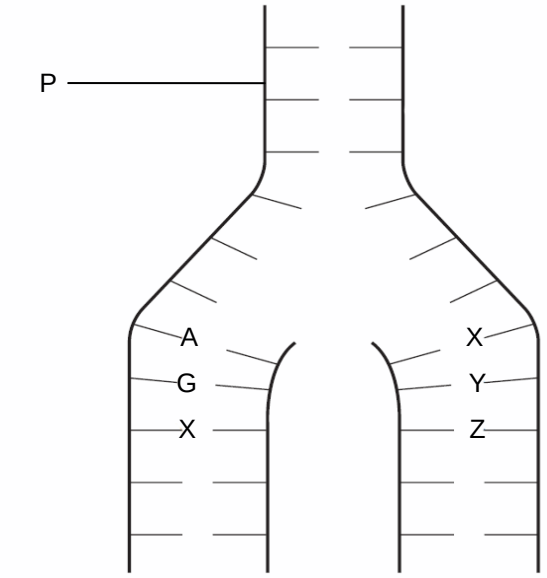
A. fertilization of an ovum by two sperms
B. non-disjunction of sex chromosomes
C. having 21 pairs of chromosomes in each body cell
D. having three copies of chromosome 21 in each body cell ☐

Question	1	2	3	4	5	6	7	8	9	10
Related section	27.1	27.1	27.1	27.1	27.1	27.1	27.1	27.2	27.2	27.2

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B. Problem solving (21 marks)

11. The diagram below shows the process of DNA replication. The horizontal lines represent the positions of nucleotide bases.



(a) Which part of a DNA molecule is represented by P in the diagram? (1 mark)

(b) In the diagram, A represents the base adenine and G represents the base guanine. Name the base at positions X, Y and Z. (3 marks)

(c) Describe ***one*** way in which the process shown in the diagram differs from transcription. (1 mark)

(Total: 5 marks)

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12. (a) Some of the events which take place in protein synthesis are listed below.

A A ribosome attaches to the mRNA molecule.

B The mRNA molecule leaves the nucleus.

C The two strands of a DNA molecule separate.

D Peptide bonds forms between the amino acids.

E mRNA nucleotides join with exposed DNA bases and form a molecule of mRNA.

F tRNA molecules bring specific amino acids to the mRNA molecule.

(i) Write the letters in the correct order to show the sequence of events during protein synthesis. (2 marks)

_____ → _____ → _____ → _____ → _____ → _____

(ii) In which structure of a cell does event C take place? (1 mark)

(iii) Which of events A to F are involved in translation? (1 mark)

(b) The table below shows the mRNA codons for some amino acids.

Amino acid	mRNA codon
Valine	GUG
Alanine	GCG
Glutamic acid	GAG
Tyrosin	UAU
Cysteine	UGU

(i) A tRNA molecule has the anticodon AUA. Which amino acid does the tRNA molecule carry? (1 mark)

(ii) Give the DNA sequence coding for cysteine. (1 mark)

(Total: 6 marks)

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★ 13. Lysozyme is an enzyme consisting of a single polypeptide chain of 129 amino acids.

(a) What is the minimum number of nucleotide bases needed to code for lysozyme? (1 mark)

(b) A section of mRNA sequence of lysozyme is shown below.

GGUCUUUCUUAUGGUAGAUAU

(i) Give the base sequece of DNA from which this section of mRNA is transcribed. (1 mark)

(ii) How many different types of tRNA molecule would attach to this section of mRNA? (1 mark)

(c) Gene mutations occur naturally. Give ***two*** factors that might increase the frequency at which gene mutations occur. (2 marks)

(d) Two single base mutations occurred in the DNA coding for this section of mRNA. These mutations caused an alteration in the sequence of amino acids in lysozyme. The original and altered sequences of amino acids are shown below.

Original sequence: Gly – Leu – Ser – Tyr – Gly – Arg – Tyr

Mutated sequence: Gly – Leu – Tyr – Leu – Trp – Arg – Tyr

The table below shows the mRNA codons for some amino acids.

Amino acid	**mRNA codon**
Arg	AGA
Gly	GGU
Leu	CUU or UUA
Ser	UCU
Trp	UGG
Tyr	UAU or UAC

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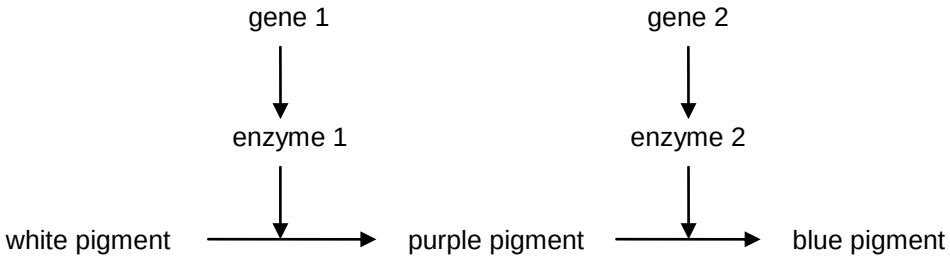
(i) Using the information provided, deduce the base sequence of the mutated mRNA.
(1 mark)

(ii) Determine the nature and location of the two single base mutations in the DNA.
(4 marks)

(Total: 10 marks)

C. Exam challenge (9 marks)

★★ 14. Flax plants normally produce blue flowers and some mutants produce white or purple flowers. The diagram below shows the pathway for producing the coloured pigments found in flax flowers. Genes 1 and 2 code for enzymes 1 and 2 respective. These two enzymes catalysed the conversion of one coloured pigment to another.



(a) A deletion mutation occurs in gene 1. Describe how a deletion mutation affects a gene.
(2 marks)

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(b) Describe and explain how the altered gene could result in flax plants with white-coloured flowers. (4 marks)

(c) In order to separate the enzymes involved in the above pathway, a scientist used the extracts from flax petals of different colours to carry out electrophoresis. The results are shown in the table below. Each band represents a functional enzyme.

Based on the result, deduce the colour of petals in each case. (3 marks)

Result of electrophoresis	**Colour of petal**
	White
	(i)
	(ii)
	(iii)

(Total: 9 marks)

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Marking scheme

A. Concept checking (10 marks, 1 mark each)

1.	C	2.	B	3.	A	4.	D	5.	B
6.	C	7.	A	8.	B	9.	A	10.	D

(1 mark each)

B. Problem solving (21 marks)

11. (a) The sugar phosphate backbone (1)

(b) X: base T / thymine (1)

Y: base C / cytosine (1)

Z: base A / adenine (1)

(c) During transcription, only one of the DNA strands serve as the template. (1)

(Total: 5 marks)

12. (a) (i) C → E → B → A → F → D (2)

(4 correct = 1 mark; less than 4 correct = 0)

(ii) nucleus (1)

(iii) A, D, F (1)

(b) (i) Tyrosin (1)

(ii) ACA (1)

(Total: 6 marks)

13. (a) 390 (1)

(b) (i) CCAGAAAGAATACCATCTATA (1)

(ii) 5 (1)

(c) Any ***two*** of the following: (2)

- exposure to high energy radiations / X rays / ultraviolet light / gamma rays (1)
- exposure to chemical mutagens (e.g. nitrite / tar / PAH / asbestos / dioxin) (1)
- expose to a mutagen for a long time (1)
- expose to high dosage of mutagen (1)

(d) (i) GGUCUUUACUUAUGGAGAUAU (1)

(ii) There is an addition of thymine (1)

after the 7th base of the original DNA sequence (1)

and a deletion of adenine (1)

after the 14th base of the original DNA sequence. (1)

(Total: 10 marks)

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C. Exam challenge (9 marks)

14. (a) A deletion of bases by a number other than a multiple of three (1)
will change the grouping of bases / cause a shift of the reading frame. (1)

(b) The reading frame shift causes every mRNA codon after the point of mutation to change. (1)

This results in a polypeptide with a different sequence of amino acids. (1)

The polypeptide folds into a protein with a different conformation. (1)

The protein (enzyme 1) is non-functional. (1)

As a result, white pigment cannot be converted to purple and blue pigments.

(c) (i) blue (1)

(ii) purple (1)

(iii) white (1)

(Total: 9 marks)

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